

MISRATA, Libyan Arab JamahiriyaWMO: **620160**Lat: **32.325N**Lon: **15.059E**Elev: **108**StdP: **14.64**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	46.8	48.6	27.9	21.9	63.9	31.6	26.0	61.7	34.9	57.8	33.2	56.9	7.5	240	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	98.9	71.3	94.6	70.9	91.1	71.2	79.7	85.2	78.6	84.5	77.7	83.9	13.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.1	146.8	83.2	77.0	141.1	82.6	75.9	135.7	82.1	43.3	85.4	42.2	84.5	41.2	84.0	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.6	24.2	21.2	DB	43.0	108.6	3.8	4.4	40.3	111.7	38.1	114.3	36.0	116.7	33.2	119.9	(4)
(5)			WB	37.7	84.0	2.3	3.1	36.1	86.2	34.8	88.0	33.5	89.7	31.9	92.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.1	58.8	58.4	61.7	66.1	71.8	77.2	80.9	82.6	80.8	74.8	66.7	60.3	(6)	
	DBStd	9.84	3.18	4.53	5.23	5.38	5.84	5.24	3.75	2.84	3.40	4.58	4.44	4.29	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	757	194	196	133	43	4	0	0	0	0	0	29	158	(9)	
	CDD50	7326	273	234	362	482	677	817	959	1010	924	767	502	319	(10)	
	CDD65	2611	3	10	31	75	216	367	494	545	474	303	81	12	(11)	
	CDH74	23247	13	63	232	473	1382	2933	5077	6183	4628	1913	302	48	(12)	
(13)	CDH80	8266	0	16	76	177	568	1170	1813	2365	1497	519	59	6	(13)	
(14)	Wind	WSAvg	10.2	10.6	11.4	11.4	11.6	11.1	9.8	9.2	8.7	9.5	9.3	10.4	(14)	
(15) Precipitation	PrecAvg	10.6	2.2	1.0	0.8	0.4	0.2	0.0	0.0	0.0	0.4	1.6	1.6	2.2	(15)	
	PrecMax	16.9	8.5	2.8	2.9	2.1	1.5	0.7	0.0	0.7	2.6	5.4	7.9	6.5	(16)	
	PrecMin	3.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)	
	PrecStd	3.3	1.9	0.7	0.8	0.4	0.3	0.1	0.0	0.2	0.6	1.5	1.5	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.9	81.4	88.6	95.0	100.3	106.0	102.6	101.0	101.6	95.5	86.3	79.6	(19)	
		MCWB	55.8	58.2	60.1	61.9	68.7	71.0	71.8	73.0	71.2	67.2	62.4	57.4	(20)	
	2%	DB	70.2	73.8	79.8	85.1	91.7	96.1	96.5	95.0	94.2	90.0	80.7	73.2	(21)	
		MCWB	55.2	56.1	58.6	63.2	66.3	69.6	72.0	73.9	72.4	67.5	63.4	57.5	(22)	
	5%	DB	67.1	69.0	74.4	79.5	85.6	90.7	91.5	91.3	89.8	85.3	77.1	69.6	(23)	
		MCWB	55.9	55.9	58.4	61.0	66.1	69.2	72.4	74.4	73.7	69.0	63.4	57.9	(24)	
	10%	DB	64.8	65.3	69.9	74.5	80.8	86.6	87.5	88.4	86.7	81.8	74.5	67.2	(25)	
		MCWB	56.1	55.6	57.7	60.7	65.0	69.3	73.4	75.4	74.7	70.1	63.9	57.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.7	61.1	64.2	74.6	73.5	77.7	80.2	82.1	81.0	76.6	70.9	64.1	(27)	
		MCDB	65.6	69.8	73.7	86.4	90.0	84.1	85.3	88.8	85.5	83.1	75.9	69.9	(28)	
	2%	WB	60.2	59.6	62.4	66.1	70.2	75.5	78.8	79.9	79.2	74.5	68.4	62.0	(29)	
		MCDB	64.8	66.7	69.5	74.9	80.5	82.1	84.2	85.5	84.5	80.5	74.9	67.6	(30)	
	5%	WB	58.7	58.5	61.2	64.2	68.9	74.3	77.8	78.9	78.0	73.1	66.9	60.8	(31)	
		MCDB	64.0	64.9	68.4	71.9	77.9	81.1	83.6	84.7	83.7	79.5	73.8	66.3	(32)	
	10%	WB	57.4	57.4	60.0	62.9	67.6	73.0	76.8	78.0	76.8	71.8	65.5	59.4	(33)	
		MCDB	62.6	63.8	67.0	70.3	75.5	80.1	83.0	84.2	83.0	79.0	72.4	65.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	13.2	14.2	13.5	14.1	14.3	12.5	12.4	12.3	13.3	13.7	12.9	(35)	
		MCDBR	17.5	20.8	23.8	24.9	24.3	24.0	21.6	19.2	19.7	19.6	18.0	17.2	(36)	
	5% WB	MCWBR	9.6	9.6	9.5	9.3	7.4	7.0	7.8	6.9	7.4	7.5	9.3	10.3	(37)	
		MCWBR	12.3	14.7	16.3	15.0	16.7	14.4	12.7	11.6	12.2	13.5	14.1	12.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.413	0.485	0.540	0.542	0.481	0.479	0.479	0.516	0.472	0.427	0.394	(40)	
		taud	2.231	2.139	1.937	1.832	1.854	2.042	2.066	2.082	1.980	2.065	2.137	2.211	(41)	
	Ebn at Noon		258	263	252	243	244	259	258	256	240	240	240	247	(42)	
		Edh at Noon	36	43	57	66	66	54	53	51	54	46	39	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		947	1207	1578	1923	2147	2328	2335	2107	1727	1319	1012	828	(44)	
	RadStd		51	66	79	73	92	82	36	54	58	70	40	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page